PULLING A RELATIVE CYCLE INTO THE IMAGE OF THE CYCLE-CLASSES MAP BY A COMPLETELY DECOMPOSED PROPER COVER

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Allowed standard prerequisites (editorial).

- Completely decomposed morphisms (More on Morphisms, Definition more-morphisms-definition-cd-morphism, canonical tag 0GTI). cited at statement lines 1180-1181. The class of morphisms g : S' to S required in the statement.
- The cycle-classes construction (relative-cycles chapter, Equation equation-cycle-classes, canonical tag 0H58). cited at statement line 1183 and proof lines 1220 and 1229. The image of which g^*alpha must be brought.
- Generic flatness (Morphisms, Proposition morphisms-proposition-generic-flatness, canonical tag 052A). cited at proof lines 1213-1215. Makes the closures Z_i flat over a nonempty open U of the integral base.
- Flatness after blowing up (More on Flatness, Lemma flat-lemma-flat-after-blowing-up, canonical tag 081R). cited at proof lines 1216-1218. Provides the U-admissible blowing up g : S' to S on which the strict transforms Z'_i are flat over S'.
- Uniqueness of extensions of relative cycles agreeing at the generic point (relative-cycles chapter, Lemma lemma-uniqueness-extension, canonical tag 0H55). cited at proof lines 1220-1221. Identifies beta = sum n_i [Z'_i/X_{S'}/S']_r with g^*alpha.

Source reference. Original definition 0GTI: sources/stacks-chow/tags.txt, source lines 20968–20968. Cited at statement lines 1180-1181: the morphism class required in the statement.

Source reference. Original equation 0H58: sources/stacks-chow/tags.txt, source lines 21378–21378. Cited at statement line 1183 and proof lines 1220 and 1229: the image of which g^*alpha is brought.

Source reference. Original proposition 052A: sources/stacks-chow/tags.txt, source lines 6563–6563. Cited at proof lines 1213-1215: flatness of the closures over a nonempty open.

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 081R: sources/stacks-chow/tags.txt, source lines 10226–10226. Cited at proof lines 1216-1218: the U-admissible blowing up g : S' to S.

Source reference. Original lemma 0H55: sources/stacks-chow/tags.txt, source lines 21375–21375. Cited at proof lines 1220-1221: identifies beta with g^*alpha.

Lemma 0.2. *Let $f : X \to S$ be a finite type morphism of schemes with S Noetherian. Let $r \geq 0$. Let α be a relative r-cycle on X/S. Then there is a proper, completely decomposed (More on Morphisms, Definition 0GTI) morphism $g : S' \to S$ such that $g^*\alpha$ is in the image of (0H58).*

Proof. By Noetherian induction, we may assume the result holds for the pullback of α by any closed immersion $g : S' \to S$ which is not an isomorphism.

Let $S_1 \subset S$ be an irreducible component (viewed as an integral closed subscheme). Let $S_2 \subset S$ be the closure of the complement of S' (viewed as a reduced closed subscheme). If $S_2 \neq \emptyset$, then the result holds for the pullback of α by $S_1 \to S$ and $S_2 \to S$. If $g_1 : S'_1 \to S_1$ and $g_2 : S'_2 \to S_2$ are the corresponding completely decomposed proper morphisms, then $S' = S'_1 \amalg S'_2 \to S$ is a completely decomposed proper morphism and we see the result holds for S[1] . Thus we may assume $S' \to S$ is bijective and we reduce to the case described in the next paragraph.

Assume S is integral. Let $\eta \in S$ be the generic point and let $K = \kappa(\eta)$ be the function field of S. Then α_η is an r-cycle on X_K. Write $\alpha_\eta = \sum n_i[Y_i]$. Taking the closure of Y_i we obtain integral closed subschemes $Z_i \subset X$ whose base change to η is Y_i. By generic flatness (for example Morphisms, Proposition 052A), we see that Z_i is flat over a nonempty open U of S for each i. Applying More on Flatness, Lemma 081R we can find a U-admissible blowing up $g : S' \to S$ such that the strict transform $Z'_i \subset X_{S'}$ of Z_i is flat over S'. Then $\beta = \sum n_i[Z'_i/X_{S'}/S']_r$ is in the image of (0H58) and $\beta = g^*\alpha$ by Lemma 0H55.

However, this does not finish the proof as $S' \to S$ may not be completely decomposed. This is easily fixed: denoting $T \subset S$ the complement of U (viewed as a closed subscheme), by Noetherian induction we can find a completely decomposed proper morphism $T' \to T$ such that $(T' \to S)^*\alpha$ is in the image of (0H58). Then $S' \amalg T' \to S$ does the job. □

[1]Namely, any closed subscheme of $S'_1 \times_S X$ flat and of relative dimension $\leq r$ over S'_1 may be viewed as a closed subscheme of $S' \times_S X$ flat and of relative dimension $\leq r$ over S'.